Baoxi Huang

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EDUCATION

The University of Tokyo, Japan *Oct. 2024 – Sep. 2026*
Master of Environment Systems, Mizuno Lab (Machine Learning Focus)
Teaching assistant in Data Structure and Algorithm

Jilin University, China *Oct. 2020 – Jul. 2024*
Bachelor of Data Science and Big Data Technology
Honors: Top 3% of graduates / First prize in college scholarship / Third prize in Mathematical Contest in Modeling

INTERNSHIP

Dataset Deduplication for LLMs Using the LSHBloom Algorithm [Blog]
Machine Learning Engineer | Preferred Networks, Inc.
Skills: Python / Git / AWS / Distributed systems *Aug. 2025 – Sep. 2025*

Main Contribution

- Translated relevant research into a production implementation of LSHBloom and integrated it into the in-house data deduplication pipeline, resolving issues of out-of-memory and excessive processing time in the existing workflow.

Tasks

- Benchmarked LSHBloom against MinHashLSH on a self-built synthetic dataset, achieving comparable F1 (**0.6963** vs. **0.7015**).
- Deployed LSHBloom on AWS using S3, AWS Batch, and EC2 to deduplicate the peS2o dataset, reducing processing time by **38.4%** and memory usage by **87.2%**; authored a technical blog post on this work for the official Preferred Networks Tech Blog.

Extended work [GitHub]

- Built an LSHBloom deduplication pipeline for Qwen2.5-0.5B continued pretraining on peS2o, cutting training tokens by **8.46%** and A100 GPU time by **8.47%**, with validation perplexity within **0.34%** of raw data and a **1.1%** gain in SciQ normalized accuracy.

Development of Internal HR Management System Using React [Product Website]
Software Engineer | Money Forward, Inc.
Skills: React / Git / GraphQL / Github Action *Mar. 2026 – Apr. 2026*

Main Contribution

- Contributed to migrating the in-house HR management system from Vue 2 to React to improve scalability and developer productivity, delivering **5 merged pull requests within 10 days**, with contributed features deployed to production.

Tasks

- Collaborated closely with a UI engineer to implement the employee payroll page from Figma designs, displaying monthly salaries, bonuses, and reimbursements for each employee, and wrote unit tests to verify the functionality and correctness of the page.
- Enabled the side-bar feature and implemented the GraphQL API for the employee-information workflow form submissions.

Extended work [GitHub]

- Built a local-first HR onboarding platform (React / TypeScript / Node) in which HR composes three-section templates, invites employees and reviews submissions, supporting draft saving, account-number masking, a notification centre and an audit trail.
- Implemented a pluggable tool-calling AI agent that performs the HR work from Japanese instructions through **5 guarded tools** that reuse the UI's own domain validation, so no agent action can bypass validation or ownership checks, with a deterministic offline provider, an OpenAI-compatible (Deepseek) backend and a scheduled job chasing non-submitters.

PROJECT

SupportMesh: Multi-Agent Customer Support Platform [GitHub]
Skills: Python / FastAPI / Redis / ChromaDB / Docker / Prometheus *Apr. 2026 – Present*

Overview

- Built an **explainable, observable multi-agent support harness** for enterprise software that routes product inquiries, technical incidents, and after-sales requests across **3 agents**, with human escalation, Prometheus monitoring, and Docker deployment.

Core Components

- Designed a hybrid intent router fusing LLM, embedding, and pattern-matching signals to classify requests into **19 intents** and **4 urgency levels**, reaching **91.3% accuracy** and **0.89 macro-F1** on about **500 labeled cases** (per-class F1: **0.87–0.92**).
- Built **intent-gated ChromaDB RAG** over an enterprise knowledge base with 3 query rewrites, multi-path retrieval, deduplication, reranking, caching, and circuit-breaker fallback, achieving **88.7% answer accuracy** and **91.2% retrieval recall**.
- Implemented Redis + ChromaDB-based **tiered memory**, dynamic Skills injection, monitor-driven route down-weighting, and **LLM-as-Judge** evaluation, with a 24-hour TTL, compression after 15 messages retaining the latest 5, hot loading for 3 Skill categories, 5 end-to-end suites, and 5% degradation alerts across customer-support workflows.

PUBLICATION & CERTIFICATION

Publication: Seafloor Debris Detection Using Underwater Images and Deep Learning-Driven Image Restoration. [Paper]
Certifications: AWS Certified Solutions Architect – Associate (SAA) / TOEFL (91) / JLPT (N2)